

EN NO: 8

DATE:

MINI PROJECT HEALTHCARE DATA INSIGHTS

AIM:

To analyze Indian healthcare data using **MySQL and Power BI** by managing a 40,000-record healthcare dataset, performing data analysis using SQL queries, and creating interactive visualizations to identify trends in patient demographics, diseases, medical conditions, treatments, treatment outcomes, BMI, blood glucose, HbA1c levels, regions, and insurance coverage.

Software and Tools Required

- MySQL
- MySQL Workbench
- Power BI Desktop

Procedure

Step 1 — Project Setup

The Indian healthcare dataset containing exactly **40,000 patient records** was obtained in CSV format and prepared for analysis.

The project uses the following structure:

HealthcareDataInsights

```
|
|
|— indian_healthcare_40000.csv
|
|
|— MySQL Database
|   └─ healthcare_analytics
|       └─ indian_healthcare_40000
|
|
└─ Power BI Dashboard
```

Step 2 — Create Database

Open **MySQL Workbench** and connect to the MySQL server.

Create the database:

```
CREATE DATABASE healthcare_analytics;
```

Select the database:

```
USE healthcare_analytics;
```

Step 3 — Import / Create Table

The healthcare dataset was imported into MySQL and stored in the table:

indian_healthcare_40000

The table contains the following attributes:

Attribute	Description
Patient_ID	Unique patient identifier
Age	Age of the patient
Gender	Gender of the patient
Region	Region/state
Socioeconomic_Status	Socioeconomic category
Occupation	Patient occupation
Primary_Diagnosis	Primary medical diagnosis
Symptoms	Patient symptoms
Blood_Glucose_mg_dL	Blood glucose level
HbA1c_%	HbA1c percentage
BMI	Body Mass Index
Hospital_Type	Type of hospital
Imaging_Type	Type of imaging
Imaging_Findings	Imaging findings
Insurance_Covered	Insurance coverage status
Treatment_Type	Type of treatment
Total_Cholesterol_mg_dL	Total cholesterol level

Step 4 — Verify the Dataset

The number of records in the table was verified using SQL.

Run:

USE healthcare_analytics;

```
SELECT COUNT(*) AS total_records
```

```
FROM indian_healthcare_40000;
```

Result:

```
total_records
```

```
-----
```

```
40000
```

This confirms that all **40,000 healthcare records** have been successfully stored in MySQL.

Step 5 — Data Analysis Using MySQL

SQL queries were used to analyze the healthcare dataset.

The following analyses were performed:

- Gender distribution
- Age group distribution
- Region-wise patient distribution
- Socioeconomic status
- Primary diagnosis
- Blood glucose levels
- HbA1c levels
- BMI distribution
- Treatment types
- Treatment outcomes
- Insurance coverage
- Diagnosis versus treatment outcome

The data flow is:

Healthcare CSV Dataset



MySQL Server



healthcare_analytics



indian_healthcare_40000



SQL Analysis



Power BI



Interactive Dashboard

Data Visualization Using Power BI

After performing the data analysis using MySQL, the healthcare dataset was imported into **Power BI Desktop**.

Different visualizations were created to represent healthcare trends and patterns.

Visualization 1 — Patients by Primary Diagnosis

Chart: Bar Chart

Shows the number of patients for each primary diagnosis.

This visualization helps identify the most common diseases in the dataset.

Visualization 2 — Patients by Region

Chart: Bar Chart

Shows the number of patients across different regions.

Tamil Nadu has the highest patient count.

Visualization 3 — Gender Distribution

Chart: Donut Chart

Shows the distribution of patients based on gender:

- Male
- Female
- Other

Visualization 4 — Age Group Distribution

Chart: Column Chart

Shows the distribution of patients across different age groups:

- 0–17
- 18–30
- 31–45
- 46–60
- 61+

Visualization 5 — BMI Distribution

Chart: Column Chart

Shows the distribution of patients according to BMI categories:

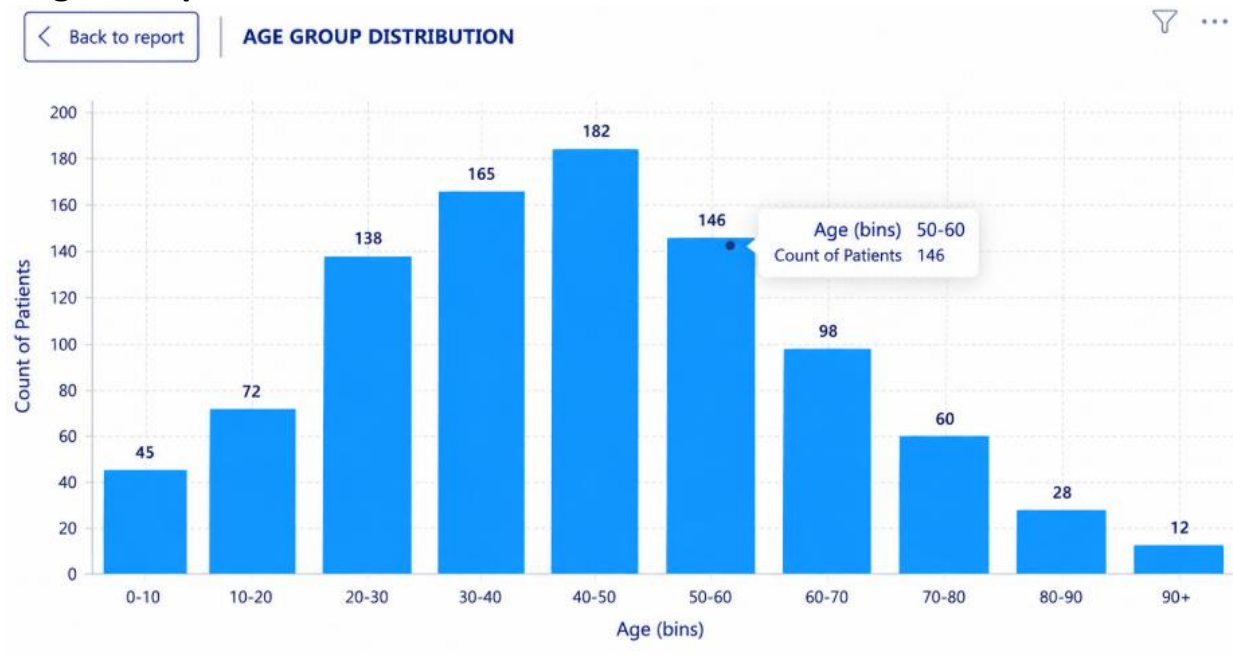
- Underweight
- Normal
- Overweight
- Obese

OUTPUT:

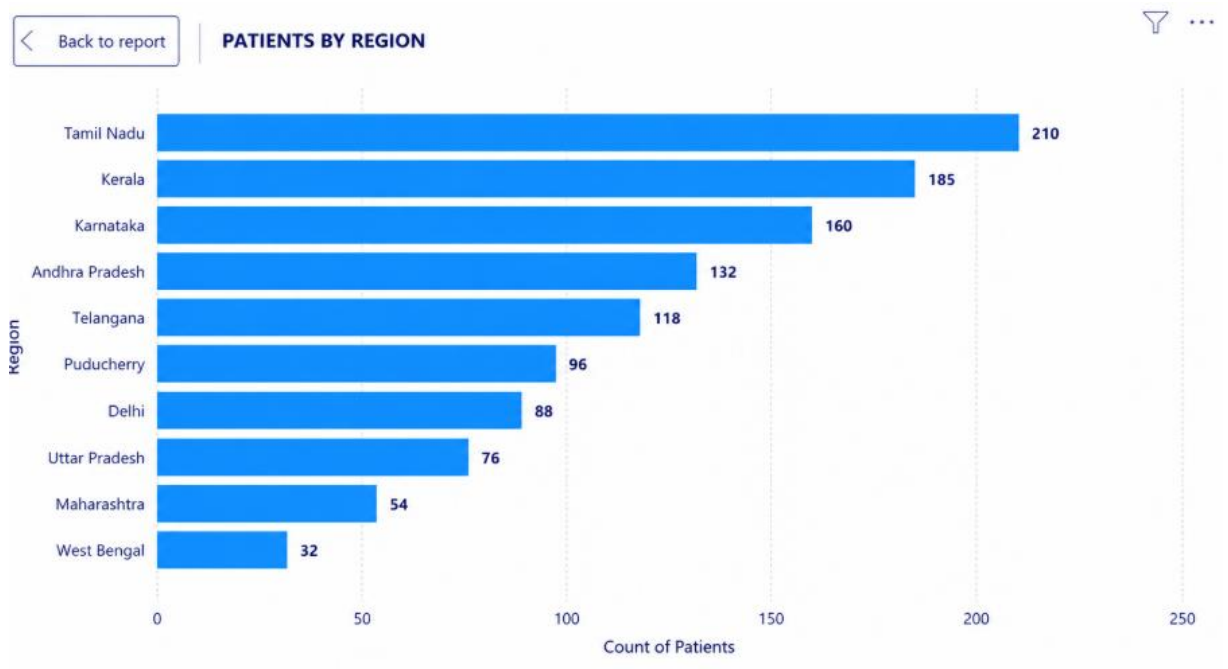
Gender Distribution



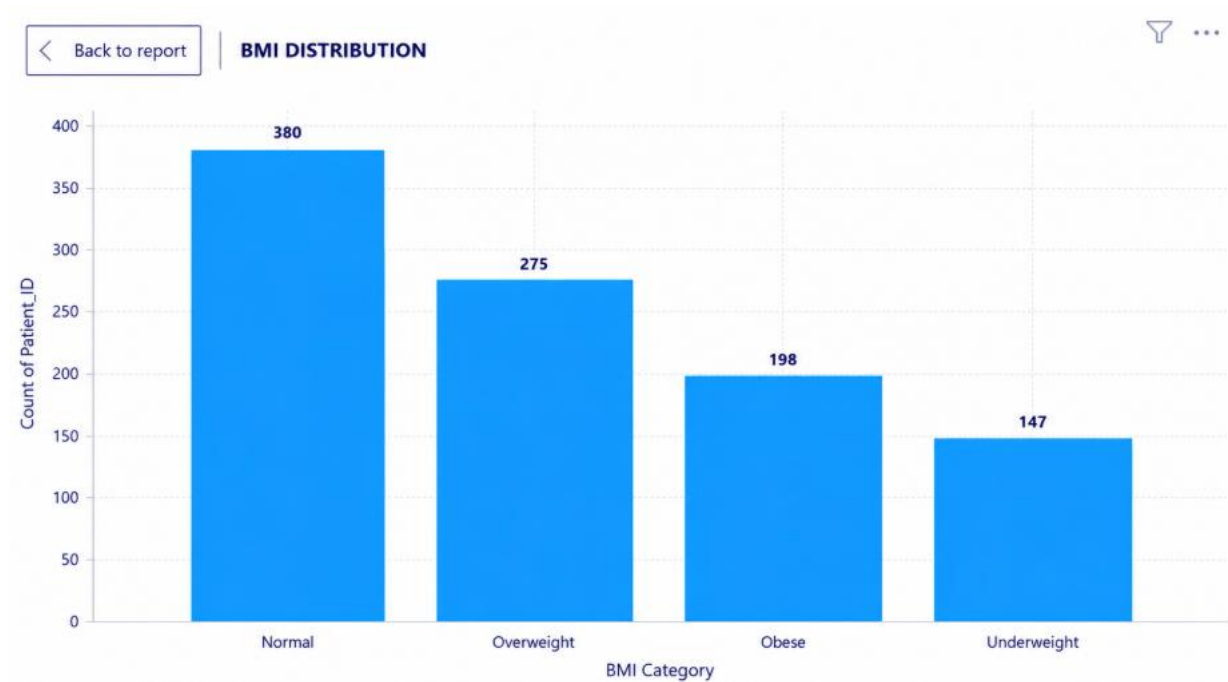
Age Group Distribution



Region-wise Patient Distribution



BMI Distribution



```

1 CREATE DATABASE healthcare_analytics;
2 USE healthcare_analytics;
3 SELECT COUNT(*) AS total_records
4 FROM healthcare_patients;
5 USE healthcare_analytics;
6 SELECT DATABASE();
7 SHOW TABLES;
8 USE healthcare_analytics;
9
10 SELECT DATABASE();
11
12 SHOW TABLES;
13
14 USE healthcare_analytics;
15
16 SELECT COUNT(*) AS total_records
17 FROM indian_healthcare_20000;
18
19

```

Patient_ID	Age	Gender	Region	Socioeconomic_Status	Occupation	Visit_Date	Symptoms	Primary_Diagnosis	Blood_Glucose_mg...
92CF3498-6D6	46	Female	Tamil Nadu	Low	Engineer, control and instrumen...	2024-06-10	Headache, joint pain, high fever	Cardiovascular Disease	122.9
896C7353-4EF	50	Male	West Bengal	Middle	Insurance underwriter	2024-09-10	Swelling in legs, high BP	Type 2 Diabetes	119

#	Time	Action	Message	Duration / Fetch
50	17:11:50	SELECT Treatment_Outcome, COUNT(*) AS Patient_Count FROM indian_healthcare_20000 GROUP BY Treatm...	5 row(s) returned	0.032 sec / 0.000 sec
51	17:12:05	SELECT Treatment_Outcome, COUNT(*) AS Patient_Count FROM indian_healthcare_20000 GROUP BY Treatm...	5 row(s) returned	0.031 sec / 0.000 sec
52	17:12:27	SELECT Insurance_Covered, COUNT(*) AS Patient_Count FROM indian_healthcare_20000 GROUP BY Insuran...	2 row(s) returned	0.015 sec / 0.000 sec
53	17:13:17	SELECT Primary_Diagnosis, Treatment_Outcome, COUNT(*) AS Patient_Count FROM indian_healthcare_200...	50 row(s) returned	0.047 sec / 0.000 sec
54	17:14:38	USE healthcare_analytics	0 row(s) affected	0.000 sec
55	17:14:38	SELECT * FROM indian_healthcare_20000 LIMIT 0, 1000	1000 row(s) returned	0.000 sec / 0.000 sec

RESULT:

```

61 FROM indian_healthcare_20000;
62
63 SELECT
64 CASE
65 WHEN Age < 18 THEN '0-17'
66 WHEN Age BETWEEN 18 AND 30 THEN '18-30'
67 WHEN Age BETWEEN 31 AND 45 THEN '31-45'
68 WHEN Age BETWEEN 46 AND 60 THEN '46-60'
69 ELSE '61+'
70 END AS Age_Group,
71 COUNT(*) AS Patient_Count,
72 ROUND(
73 COUNT(*) * 100.0 /
74 (SELECT COUNT(*) FROM indian_healthcare_20000),
75 2) AS Percentage
76 FROM indian_healthcare_20000
77 GROUP BY Age_Group
78
79

```

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Thus, the healthcare data was successfully analyzed using **MySQL** and visualized using **Power BI**. The project helped to identify patterns in patient demographics, diseases, treatments, treatment outcomes, BMI, and insurance coverage. The interactive dashboard provides a simple and clear way to understand the healthcare data.